"ATC FOR ENGINE BLOCK ASSEMBLY USING ABB IRB 1200 ROBOT"

PROJECT REPORT

Submitted by

NIKHIL JOHN CL	**(71812210028)**
PRANESH E	**(71812210035)**
SMITH V JOSEPH	**(71812210045)**

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Vattamalaipalayam, N.G.G.O. Colony Post,
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Certified that this project report entitled **"ATC for Engine Block Assembly using ABB IRB 1200 Robot"** is the bonafide work of **NIKHIL JOHN CL (71812210028), PRANESH E (71812210035), SMITH V JOSEPH (71812210045),,** who carried out the **20RA280** Project Work under my supervision.

SIGNATURE	**SIGNATURE**
SUPERVISOR	**HEAD OF THE DEPARTMENT**
Dr. A .KISHORE KUMAR	**Dr. A. MURUGARAJAN,**
Associate Professor	Professor and Head
Department of Robotics and Automation	Department of Robotics and Automation
Ramakrishna Engineering College	Sri Ramakrishna Engineering College
Coimbatore - 641022	Coimbatore – 641022

Submitted for Project Viva Voce Examination held on _________________

Internal Examiner **External Examiner**

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ABSTRACT

The automation of manufacturing processes has become essential in improving productivity, precision, and safety in modern industries. This study focuses on the design and implementation of an **Automatic Tool Changer (ATC)** for engine block assembly using the ABB IRB 1200 robot. Engine block assembly involves multiple operations such as fastening, positioning, inspection, and material handling, which traditionally require different tools and manual intervention. The integration of an ATC system enables the robot to automatically switch between various tools, thereby reducing cycle time and increasing operational efficiency.

The proposed system utilizes the flexibility and high precision of the ABB IRB 1200 robot, combined with a tool-changing mechanism that allows seamless transitions between end-effectors such as grippers, screwdrivers, and inspection sensors. The control system is programmed to identify task requirements and execute tool changes without human assistance. This enhances production speed while maintaining consistent quality standards.

In addition, the implementation of ATC minimizes downtime associated with manual tool replacement and reduces the risk of human error. The system also improves workplace safety by limiting direct human interaction with heavy machinery. The compact design and high repeatability of the robot make it particularly suitable for confined assembly environments. The results of this study demonstrate that the integration of an ATC system with robotic automation significantly improves productivity, flexibility, and reliability in engine block assembly processes. This approach supports the advancement of smart manufacturing and Industry 4.0 by enabling adaptive and efficient production systems.

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LIST OF ABBREVIATIONS

ATC	–	Automatic Tool Changer
BECU	–	Beryllium Copper
BLDC	–	Brushless Direct Current
CAD	–	Computer-Aided Design
DI/DO	–	Digital Input / Digital Output
DOF	–	Degrees of Freedom
EMF	–	Electromotive Force
FoS	–	Factor of Safety
HMI	–	Human Machine Interface
IRC5	–	ABB Integrated Robot Controller 5
ISO	–	International Organisation for Standardisation
MMF	–	Magnetomotive Force
MVS	–	Motion-Visual Signature
NPN	–	Negative-Positive-Negative (transistor type)
PLC	–	Programmable Logic Controller
PSU	–	Power Supply Unit

CHAPTER-1

INTRODUCTION

Industrial automation has become a fundamental part of modern manufacturing, especially in the automotive sector where high precision, consistency, and productivity are essential. With increasing demand for faster production and reduced human intervention, industries are shifting towards robotic systems to handle complex assembly operations efficiently.

Engine block assembly is a critical process that involves multiple tasks such as lifting, positioning, fastening, and inspection. Traditionally, these tasks require different tools and frequent manual intervention, which increases cycle time and reduces overall efficiency. Industrial robots provide a reliable solution; however, the use of a single end-effector limits their flexibility and capability to perform multiple operations.

To overcome this limitation, Automatic Tool Changer (ATC) systems are introduced. An ATC enables a robot to automatically switch between different tools during operation, allowing it to perform various tasks without human involvement. This significantly improves productivity, reduces downtime, and enhances the overall efficiency of the system.

In this project, an Automatic Tool Changer system is designed for engine block assembly using the ABB IRB1200 industrial robot. The system integrates multiple end-effectors, including grippers and specialized tools, enabling the robot to handle different assembly tasks seamlessly.

A major aspect of the ATC system is ensuring reliable electrical and mechanical connections during tool exchange. To achieve this, pogo pin-based electrical contacts are used for consistent power transfer, while a solenoid-based locking mechanism ensures secure and precise tool attachment.

The proposed system focuses on improving flexibility, reducing manual intervention, and enhancing operational safety. By integrating advanced tool-changing capabilities with robotic automation, this project contributes to the development of smart manufacturing systems aligned with Industry 4.0 concepts.

1.1 Project Overview (Continued)

Reliable tool engagement in an Automatic Tool Changer system requires accurate alignment between the robot flange and the tool interface. In this design, precise positioning is achieved using mechanical guiding elements such as tapered alignment pins and matching slots. These features ensure proper mating between the robot-side pogo pins and the corresponding contact pads on the tool. As a result, electrical connections are established with minimal resistance while preventing physical damage to the contact surfaces.

The high repeatability of the ABB IRB1200 industrial robot further strengthens this alignment process. Its ability to consistently return to exact positions ensures stable and repeatable tool docking across multiple tool-changing cycles, which is essential for maintaining system reliability in continuous industrial operation.

Another critical component of the system is the control strategy used for executing tool-changing operations. The robot is programmed with a structured sequence that includes approach, alignment, locking, and verification stages. During the pickup and drop-off process, the robot carefully navigates to the tool station, aligns itself using the guiding features, and activates the locking mechanism.

To enhance system safety and reliability, sensors and feedback mechanisms can be incorporated to verify successful tool engagement. These signals confirm both mechanical locking and electrical connectivity before the robot proceeds with the assigned task. This reduces the risk of tool slippage, disconnection, or operational failure. The solenoid-based locking mechanism is controlled through the robot's digital output signals, allowing precise synchronization with the robot's motion program.

The use of pogo pins for electrical connectivity introduces additional design considerations such as current handling capacity, contact force, and long-term durability. The selected pogo pins are capable of carrying the required electrical load of various end-effectors while maintaining low contact resistance. Their spring-loaded construction compensates for minor misalignments and ensures stable contact even under vibration or dynamic motion.

Furthermore, the pogo pins are typically manufactured with corrosion-resistant materials, such as gold-plated contacts, which improve conductivity and extend service life.

This makes them suitable for repeated engagement cycles in industrial environments where reliability and durability are critical.

Overall, the integration of precise mechanical alignment, intelligent control strategy, and robust electrical interfacing results in a highly efficient and dependable Automatic Tool Changer system suitable for advanced engine block assembly applications.

CHAPTER-2

LITERATURE REVIEW

2.1 INTRODUCTION

The increasing demand for automation in manufacturing has led to significant advancements in industrial robotics, particularly in the development of flexible assembly systems. One of the key innovations in this field is the integration of Automatic Tool Changers (ATCs), which enable robots to perform multiple tasks by switching between different end-effectors. This section presents an overview of existing research and technologies related to robotic tool-changing systems, electrical interfacing methods, and locking mechanisms used in automated assembly applications.

Industrial robots are widely used in manufacturing due to their precision, repeatability, and ability to operate in hazardous environments. Robots such as the ABB IRB1200 are commonly employed in assembly operations because of their compact design, high positional accuracy, and reliability. Researchers have highlighted that traditional robotic systems equipped with fixed end-effectors limit flexibility and reduce the overall efficiency of production systems. This limitation has driven the development of Automatic Tool Changer systems to enhance the versatility of robotic platforms.

Automatic Tool Changers play a crucial role in improving productivity by reducing downtime associated with manual tool replacement. These systems allow robots to seamlessly switch between tools required for different operations, such as gripping, fastening, and inspection. Traditional ATC systems primarily rely on pneumatic or purely mechanical locking mechanisms, often involving compressed air systems. While effective, these approaches can increase system complexity and maintenance requirements.

Recent research has focused on electrically actuated locking mechanisms, particularly solenoid-based systems, due to their simplicity, ease of integration, and reduced maintenance. Solenoid locks provide reliable tool engagement and quick release, making them suitable for automated environments where precise control is required.

Another major challenge identified in the literature is establishing reliable electrical connectivity between the robot and its end-effectors. Conventional methods use wired connections or connector plugs, which are susceptible to wear, misalignment, and frequent maintenance due to repeated connection cycles. To overcome these limitations, researchers have explored the use of spring-loaded electrical contacts, commonly known as pogo pins.

Pogo pins are widely used in electronic testing and modular device connections because of their reliability and ability to withstand repeated engagement cycles. Studies have shown that pogo pins provide consistent electrical connections with low contact resistance, making them suitable for integration in ATC systems. Their spring-loaded mechanism allows them to compensate for minor positional errors, ensuring stable connectivity even when perfect alignment is not achieved.

Several researchers have analyzed the performance of pogo pins based on parameters such as contact force, electrical resistance, durability, and current-carrying capacity. It has been observed that pogo pins with gold-plated contacts offer improved conductivity and corrosion resistance, thereby enhancing long-term reliability in industrial environments.

Mechanical alignment and repeatability are also critical factors in the design of ATC systems. Various alignment techniques, including guide pins, tapered slots, and kinematic coupling mechanisms, have been proposed to ensure accurate positioning between the robot and the tool interface. The high repeatability of industrial robots ensures that these alignment features function effectively, minimizing wear and preventing damage to electrical contacts.

In addition, researchers have introduced compliance mechanisms to accommodate minor positional inaccuracies during the docking process. These mechanisms improve tool engagement reliability and reduce the risk of mechanical stress or misalignment during repeated operations.

Overall, the literature indicates that the combination of advanced locking mechanisms, reliable electrical interfacing using pogo pins, and precise mechanical alignment techniques plays a vital role in the development of efficient and robust Automatic Tool Changer systems. These advancements contribute significantly to improving the flexibility, productivity, and reliability of robotic assembly systems in modern manufacturing environments.

2.2 BACKGROUND

The use of solenoid-based locking mechanisms in automation systems has gained attention due to their simplicity and effectiveness. Solenoids convert electrical energy into linear motion, enabling controlled locking and unlocking actions. Studies show that solenoid locks provide quick response times and can be easily integrated with programmable logic controllers (PLCs) or robot controllers. In comparison to pneumatic systems, solenoids

Eliminate the need for external air supply and reduce system complexity. However, considerations such as consumption, heat generation, and fail-safe operation must be addressed during design.

Safety and reliability are critical factors discussed in the literature related to robotic tool changers. Researchers emphasize the importance of fail-safe mechanisms to prevent accidental tool release during operation. Many systems are designed such that the locking mechanism remains engaged in the absence of power, ensuring operational safety. Additionally, sensor-based feedback systems are often incorporated to verify proper tool attachment before the robot executes its task.

Recent trends in Industry 4.0 and smart manufacturing have further influenced the development of advanced ATC systems. The integration of sensors, real-time monitoring, and intelligent control algorithms enables predictive maintenance and improved system performance. Researchers have also explored modular and reconfigurable tool changing systems that can adapt to varying production requirements.

These advancements highlight the growing importance of flexibility and scalability in modern manufacturing environments.

In summary, the literature indicates that the integration of Automatic Tool Changers significantly enhances the functionality and efficiency of industrial robots. The adoption of pogo pin technology for electrical interfacing and solenoid-based mechanisms for tool locking represents a modern and effective approach to ATC design. These technologies address key challenges related to reliability, ease of integration, and operational efficiency. The insights gained from existing research provide a strong foundation for the development of the proposed ATC system for engine block assembly.

Similarly to the essential components of robotic device changing structures, numerous studies have centered on improving the efficiency and intelligence of ATC mechanisms through optimized layout and control strategies. Researchers have emphasized that the effectiveness of an ATC gadget isn't always best dependent on mechanical and electric design but also on seamless integration with the robot manipulate structure. Superior control algorithms permit particular synchronization between robot movement and tool-changing operations, reducing cycle time and improving overall machine responsiveness.

The function of kinematic layout in device changers has also been broadly explored. Kinematic coupling mechanisms, along with 3-factor touch structures, are frequently used to make sure deterministic positioning at some stage in tool engagement. Those structures provide high repeatability and accuracy with the aid of constraining all ranges of freedom in a predictable manner. as compared to traditional flat-surface touch designs, kinematic couplings reduce positioning errors and decorate the longevity of each mechanical and electrical interfaces. While combined with the excessive repeatability of robots just like the ABB IRB1200 commercial robotic, such mechanisms appreciably improve the consistency of tool docking operations.

2.3 SUMMARY

Any other important vicinity of research is the optimization of contact interfaces for energy and sign transmission. While pogo pins are broadly diagnosed for their reliability, research have investigated various configurations to maximise their overall performance.

Parameters such as spring pressure, geometry, plating material, and vicinity are carefully selected to balance electrical conductivity and mechanical durability. Researchers have additionally explored multi-pin configurations to guide both electricity and records transmission concurrently, permitting extra advanced quit-effectors with sensors and communique skills.

Wear and fatigue evaluation of pogo pins is every other important subject matter discussed inside the literature. Due to the fact ATC structures contain frequent connection and disconnection cycles, the elements are subjected to mechanical pressure and degradation over time. Experimental research have proven that nicely designed pogo pins can face up to thousands to tens of millions of cycles without considerable lack of performance. Floor coatings which includes gold or corrosion-resistant materials further enhance durability and ensure solid resistance for the duration of the operational life.

The integration of sensing and feedback mechanisms in ATC structures has additionally received sizeable interest. Structures regularly include proximity sensors, restrict switches, or electrical continuity exams to affirm successful device attachment. Those comments structures are essential for stopping operational screw ups and ensuring that the robot executes responsibilities best whilst the device is securely established. A few advanced implementations additionally use force/torque sensors to discover peculiar situations throughout device engagement, enabling adaptive manage and error handling.

Power performance and strength control are extra factors taken into consideration in recent studies. As robotic systems emerge as greater complex, the call for for green energy distribution will increase. The usage of pogo pins facilitates lessen electricity losses related to cable connections and connector interfaces. Furthermore, researchers have explored the use of strength strategies to optimize electricity utilization, Specially in structures with a couple of quit-effectors.

From a mechanical perspective, the layout of locking mechanisms keeps to adapt with a focus on reliability and ease. At the same time as pneumatic and hydraulic systems offer high they require extra infrastructure and maintenance. In comparison, electrically actuated mechanisms along with solenoid locks offer a compact and price-effective answer.

Literature highlights that solenoid-based structures are especially appropriate for medium-load applications where fast actuation and simplicity of control are requirements. The potential to directly interface solenoids with digital control systems simplifies implementation and enhances system responsiveness.

Fail-secure layout principles also are significantly discussed within the context of ATC systems. A well-designed device changer ought to make sure that the device stays securely connected even within the occasion of electricity failure or gadget malfunction. Researchers have proposed diverse methods, inclusive of spring-loaded locking mechanisms and normally-closed solenoid systems, which preserve in the absence of electrical strength. Those designs significantly enhance safety and reliability, especially in excessive-velocity business environments.

The impact of ATC structures on production performance has been established via a couple of case research. Enforcing robotic ATCs have pronounced discounts in cycle time, improved flexibility, and improved aid usage through enabling a single robot to carry out multiple operations.

ATCs lessen the need for added robot gadgets and related expenses. This is particularly beneficial in car production, in which production strains ought to accommodate unique engine versions and frequent design changes.

Recent improvements in enterprise four.0 have in addition accelerated the abilities of ATC systems. The integration of net of factors (IoT) technology permits actual-time monitoring of tool utilization, wear, and gadget performance. Statistics collected from sensors may be analyzed to predict maintenance necessities and save you unexpected failures. Researchers have also explored the usage of synthetic intelligence and machine studying algorithms to optimize device-changing sequences and enhance choice-making in dynamic manufacturing environments.

Every other rising trend is the improvement of modular and standardized ATC interfaces. Standardization allows compatibility between specific robots and end-effectors, promoting interoperability and reducing machine integration time.

Modular designs allow clean replacement and upgrading of additives, enhancing machine scalability and adaptableness to destiny necessities.

In conclusion, the literature sincerely demonstrates that automated tool Changer structures are a important factor in contemporary robot automation. The mixture of advanced mechanical alignment techniques, dependable electric strategies consisting of pogo pins, and efficient locking mechanisms like solenoid-based systems presents a strong answer for multi-practical robotic packages.

The insights gathered from previous studies highlight the importance of reliability, precision, safety, and versatility in ATC layout. Those findings shape the muse for the improvement of the proposed system, which targets to supply progressed overall performance and efficiency in engine block assembly operations.

CHAPTER 3

PROBLEM IDENTIFICATION AND METHODOLOGY

3.1 PROBLEM IDENTIFICATION

In modern-day automobile manufacturing, engine block assembly involves multiple sequential operations which includes handling, fastening, inspection, and positioning. Every of those operations regularly calls for one of a kind types of quit-effectors, making it essential for commercial robots to interchange equipment frequently. Although robots just like the ABB IRB1200 commercial robot provide excessive precision and repeatability, their performance is confined while limited to a single give up-effector or when tool changing is performed manually.

One of the primary issues in traditional robotic systems is the shortage of a green and reliable computerized device Changer (ATC). Manual tool converting results in improved cycle time, production delays, and better dependency on human operators. This reduces the general productivity and limits the power required in cutting-edge production environments where a couple of operations must be carried out in a streamlined manner.

Any other full-size assignment is the dependable switch of electrical power and alerts among the robot and interchangeable cease-effectors. Conventional wiring strategies contain constant cables or plug-in connectors that are vulnerable to wear and tear due to repeated connection and disconnection. These techniques can also cause unfastened connections, sign loss, or electric failure. Specially in high-velocity and excessive-cycle industrial applications. Additionally, cable-based systems can restrict robotic motion and increase the complexity of the setup.

To conquer these barriers, pogo pin-based electrical touch structures present a promising opportunity.

However, integrating pogo pins into an ATC system introduces its very own set of demanding situations, including making sure proper alignment between male and lady contacts, retaining consistent contact stress, and preventing touch damage at some point of repeated operations. Any misalignment can bring about negative electric conductivity, improved resistance, or complete failure of the quit-effector.

Mechanical balance throughout device attachment is every other critical difficulty. Insufficient locking mechanisms can lead to tool misalignment, vibration, or unintentional detachment at some stage in operation. This not only influences the accuracy of meeting however additionally poses extreme protection dangers and capability harm to the work-piece or equipment. Conventional locking systems, which include guide clamps or pneumatic mechanisms, may additionally involve higher complexity, expanded upkeep, or slower response instances.

3.2 PROBLEM STATEMENT AND OBJECTIVES

Furthermore, there is a need for a compact, price-effective, and without problems controllable locking mechanism that can be seamlessly incorporated with the robotic control machine. Whilst pneumatic systems require additional infrastructure inclusive of compressors and air deliver, they will not be appropriate for all packages because of space, fee, and maintenance constraints.

Some other trouble lies in making sure repeatability and precision all through device converting. Even small positional errors can result in unsuitable docking, resulting in mechanical pressure on components or failure of electrical contact. Consequently, reaching accurate alignment among the robotic interface and the end-effector is a key venture in ATC design.

In addition to those technical challenges, protection and reliability stay fundamental concerns. The gadget ought to make certain that the tool remains securely connected for the duration of operation and does no longer detach unexpectedly, Specially in the event of electricity failure. A loss of fail-secure mechanisms can cause unsafe conditions in commercial environments.

Thinking about these demanding situations, there's a clear need for an efficient automatic device Changer gadget that:

* Permits quick and reliable device converting without guide intervention
* Provides stable and upkeep-free electric electricity transfer
* Ensures cozy and strong mechanical locking of end-effectors
* Keeps high precision and repeatability throughout tool engagement
* Enhances overall productiveness, protection, and machine flexibility

This undertaking addresses the above issues through growing an ATC device that integrates pogo pin-primarily based electric energy transfer and a solenoid door lock mechanism for comfy device attachment. The proposed answer pursuits to conquer the constraints of traditional systems even as improving the overall performance and reliability of robotic engine block assembly operations.

3.2.2 OBJECTIVES

In the modern-day technology of business automation, automobile manufacturing approaches including engine block assembly demand high precision, speed, and flexibility. Industrial robots like the ABB IRB1200 commercial robotic are broadly used due to their accuracy and repeatability. But, one of the essential boundaries in such robotic structures is the dependency on a single cease-effector or inefficient device-changing techniques. Engine block meeting entails more than one operations consisting of gripping, fastening, and inspection, every requiring specific tools. Within the absence of an efficient automatic tool Changer

(ATC), tool switching is both executed manually or through complicated mechanisms, ensuing in elevated cycle time, reduced productivity, and higher operational fees.

An essential challenge in the development of an effective ATC machine is ensuring dependable electric electricity switch among the robot and the interchangeable quit-effectors. Traditional processes depend on stressed connections or plug-in connectors, which aren't suitable for repeated engagement and disengagement due to wear, misalignment, and danger of connection failure. These boundaries can result in inconsistent overall performance, improved protection, and downtime in computerized production systems. To conquer this difficulty, pogo pin-based electrical contacts can be used as they provide a spring-loaded, self-adjusting connection. But, their implementation introduces challenges together with maintaining proper alignment, making sure regular contact stress, and stopping harm at some stage in repeated tool-converting cycles.

Similarly to electrical demanding situations, the mechanical stability of the tool attachment is a primary challenge. The end-effector have to be securely attached to the robotic to make certain precise operation and to keep away from vibrations, misalignment, or accidental detachment at some stage in excessive-velocity responsibilities. Traditional locking mechanisms, inclusive of pneumatic or manual structures, might also growth system complexity, require additional infrastructure, or lack reliability in non-stop operation. Therefore, there's a need for a compact, efficient, and without problems controllable locking gadget which could provide comfortable attachment even as allowing quick device modifications.

Another important difficulty is accomplishing high repeatability and alignment accuracy during tool engagement. Even minor positional errors can cause mistaken docking, ensuing in bad electrical touch or mechanical pressure on

components. This necessitates a well-designed interface that ensures precise alignment among the robot and the stop-effector at some point of every tool exchange cycle. Moreover, safety and reliability are crucial elements in business environments. The ATC machine need to ensure that the device remains securely locked throughout operation and need to not detach unexpectedly, Specially in case of power failure. A loss of fail-secure mechanisms can pose serious dangers to device and operators.

To address these demanding situations, this mission makes a speciality of the improvement of an automatic device Changer gadget that allows green and reliable tool switching for engine block assembly. The proposed system utilizes pogo pin-based totally electrical connections, wherein the male pins are installed on the give up-effectors and the woman contacts are included into the robot interface, making sure steady electricity switch.The primary objective of this undertaking is to layout, develop, and put into effect a strong and efficient automated device Changer (ATC) system for engine block assembly the usage of the ABB IRB1200. The system is supposed to decorate the ability, productiveness, and automation stage of manufacturing strategies by using enabling the robot to autonomously switch between more than one end-effectors during different stages of meeting. A main objective is to reap seamless and rapid tool changing without guide intervention. The ATC system is designed to decrease device alternate time and make certain excessive repeatability and precision in the course of attachment and detachment. This contributes to decreased cycle time, improved throughput, and regular assembly best in engine block operations. Any other key objective is to set up a reliable electric power transmission device among the robot and interchangeable give up-effectors. This is done the use of pogo pins, where the male connectors are established on the cease-effectors and the female connectors are incorporated into the robot-aspect interface.

Requirement	Parameter	Target Value
Gripper payload — electromagnetic	Safe hold	≥ 7.6 kg (SF ≥ 2 on 3.8 kg load)
Gripper payload — jaw (head)	Safe hold	$\geq 2\times$ head mass
ATC locking force	Axial retention	> robot max inertial force
Pogo pin current	Continuous	≤ 1.5 A per pin
Tool change cycle time	Total	< 10 seconds
Positional repeatability	Tool docking	$\pm$ 0.5 mm
Proximity sensor response	Detection time	< 5 ms

Table3.1-Requirement and Parameter

The undertaking also ambitions to ensure relaxed mechanical locking of gear to the robot flange. For this reason, a solenoid door lock mechanism is applied in the ATC device. This locking mechanism is designed to offer firm attachment, unique positioning, and vibration resistance for the duration of robot operation. It prevents unintended disengagement of gear and complements safety at some stage in excessive-speed or excessive-load meeting responsibilities.

Similarly, the venture makes a speciality of accomplishing correct alignment between the robot and the device during docking. Right alignment is vital for both mechanical coupling and electric connectivity. The system is designed to manual the cease-effector into role, ensuring reliable engagement of the locking mechanism and right touch of pogo pins.

Any other goal is to develop a compact, lightweight, and cost-effective ATC design that can be without problems incorporated with existing robotic structures. The layout emphasizes modularity, permitting distinctive forms of cease-effectors to be used for diverse operations together with gripping, fastening, or inspection in engine block assembly.

The challenge also pursuits to enhance operational protection and device reliability with the aid of incorporating fail-safe features in both electric and mechanical additives. The solenoid locking system is designed to keep device engagement even inside the occasion of strength fluctuations, while the pogo pin interface reduces the risk of electrical failure due to poor connections.

Moreover, the project seeks to reduce maintenance necessities and downtime by way of using durable additives and a simplified mechanism. Using non-permanent electrical contacts and electromechanical locking reduces wear as compared to conventional stressed connections and guide fastening structures. Eventually, the assignment objectives to illustrate the sensible implementation of a complicated ATC machine in a commercial automation scenario highlighting upgrades in efficiency, flexibility, and reliability.

3.3 METHODOLOGY AND THEORECTICAL BACKGROUND

The method for the design and implementation of the automatic tool Changer (ATC) machine for engine block meeting the use of the ABB IRB1200 entails a scientific approach protecting layout, fabrication, integration, and checking out. The system starts off evolved with a comprehensive analysis of the engine block assembly requirements, identifying the styles of quit-effectors wished, their electricity and manipulate specs, and the operational collection for the robot.

The mechanical layout of the ATC focuses on ensuring comfy and particular device attachment. A solenoid-based door lock mechanism is chosen to offer reliable engagement and release of quit-effectors. The layout procedure includes modelling the locking mechanism the use of CAD software program, simulating the mechanical masses for the duration of operation, and ensuring compatibility with the robotic flange. Right alignment publications are integrated to help in accurate docking of gear, making sure that the locking system engages efficiently with minimal manual adjustment.

For the electric power switch, the ATC makes use of pogo pin connectors, with the male pins hooked up at the give up-effectors and the woman receptacles embedded inside the robot interface. The methodology entails selecting appropriate pogo pins capable of dealing with the desired modern-day and voltage, determining the ultimate arrangement to hold strong touch, and designing the mechanical interface to make certain constant engagement even underneath dynamic robotic actions. Exact trying out is accomplished to validate electrical continuity, resistance degrees, and sturdiness of repeated mating cycles.

Integration of the ATC with the ABB IRB1200 involves each mechanical mounting and programming. The robot's end-effector interface is modified to deal with the female pogo pin meeting and the solenoid locking mechanism. The robot's controller is programmed to execute a sequence of device trade operations, together with approach, docking, locking, electrical connection verification, and undocking. Protection interlocks and sensor remarks are incorporated to save you device release at some stage in operation and to verify successful device engagement earlier than motion.

Prototyping and fabrication observe the design segment, with cautious cloth choice for sturdiness and occasional maintenance. Mechanical additives of the ATC, together with locking actuators, guiding pins, and housing, are manufactured and assembled. Electric connections are examined personally and beneath load situations. Functional trying out is conducted with the robot acting repetitive device trade cycles to assess reliability, velocity, and alignment accuracy. Any deviations are corrected by means of adjusting tolerances, spring tensions, or actuator timings.

Sooner or later, the technique includes iterative validation and optimization. The gadget is evaluated for operational efficiency, cycle time reduction, safety compliance, and simplicity of renovation. Information accrued during checking out is used to refine the design, ensuring that the ATC achieves seamless tool changes, reliable electrical connectivity thru pogo pins, and cozy tool locking via the solenoid mechanism. The methodology emphasizes modularity, permitting the ATC to conform to numerous cease-effectors for engine block assembly or different industrial applications, in the end improving automation and productiveness.

3.4 PLAN OF EXPERIMENTS

An experiment to test the effectiveness of Automated Tool Change (ATC) systems for engine block handling and assembly using ABB's IRB 1200 robotics system was set up. Tool engagement reliability, the stability of the tools when held in place and the efficiency of the operations during dynamic motion of the robot were all objectives of this study. The system consists of a 6-axis ABB IRB1200 robotic arm mounted to a robust base frame with the intent of preventing any vibration during operation.

The ATC assembly is mounted directly to the robot arm with a dual solenoid locking mechanism and mechanical engagement features; as well as a concentric alignment cone for precise tool alignment when docking to the robotic arm (robot flanges) and to the robot workspace. A male tool is mounted on the end-effector specifically to grasp an engine block weighing approximately 3.8 kg. A tool docking station is set up within the robot's work area so that tools can be picked up and released by the robot automatically without operator intervention.

The tool docking station has guide rails and alignment features to provide consistent placement of the tools exchanged with the robot. The electrical interface between the robot and ATC uses a multi-pin connector with separate lines for power and signal delivery to ensure safe operation and control of the power sent to the ATC system

The Plan of test (PoE) for the **computerized tool Changer (ATC)** machine in engine block meeting the usage of the ABB IRB1200 is designed to systematically compare mechanical reliability, electrical connectivity, operational overall performance, and safety. The experiment focuses on validating the pogo pin-primarily based energy switch, the solenoid door lock mechanism, and the general device exchange method underneath practical industrial conditions.

1. Goals:

* To verify accurate and repeatable alignment of quit-effectors with the robot interface.

* To evaluate the effectiveness of the solenoid door lock mechanism in securely fastening tools underneath static and dynamic situations.

* To assess the electrical connectivity and energy transfer efficiency of pogo pins beneath operational loads.

* To degree device change cycle time and its effect on usual meeting performance.

* To come across potential failure modes and ensure the ATC gadget operates appropriately underneath bizarre situations.

2. Experimental Setup:

* The ABB IRB1200 robotic is programmed for automatic tool docking, locking, and undocking sequences.

* Stop-effectors are ready with male pogo pins for electric energy supply; corresponding girl connectors are hooked up at the robot interface.

* Solenoid door lock mechanism is established to interact automatically upon tool docking.

* Sensors and monitoring structures measure electrical voltage/contemporary, solenoid actuation reputation, device position, and mechanical forces.

* A take a look at jig simulates engine block assembly conditions to assess tool overall performance underneath realistic load and motion.

3. *Experimental Process:*

Step 1: Alignment Verification

* Robotic techniques the tool holder at controlled pace.

* Visual markers and position sensors check for specific alignment of pogo pins and device flange.

* Misalignment tolerance is recorded and compared towards design specifications.

Step 2: Mechanical Locking test

* Solenoid door lock is actuated routinely.

* Pressure required to manually disengage the tool is measured.

* The engagement is examined underneath vibration and simulated load to verify comfy attachment.

Step 3: Electrical Connectivity check

* Energy is provided thru pogo pins.

* Voltage and contemporary stages are measured to make sure strong touch.

* Electrical continuity is examined across a couple of cycles to assess wear and reliability.

Step 4: Repetitive Cycle test

* Robotic executes multiple device alternate cycles (50–100 or extra).

* Document cycle time, alignment consistency, locking/unlocking reliability, and electric overall performance.

* Investigate mechanical components for wear and any misalignment issues.

Step five: Load overall performance test

* Quit-effector operates beneath simulated engine block assembly loads.

* Have a look at for any slippage, tool disengagement, or electric interruptions.

* Measure any deviation in tool positioning underneath dynamic conditions.

Step 6: Safety and Fault Simulation

* Simulate odd situations consisting of strength failure, misalignment, or incomplete lock engagement.

* Confirm that the ATC machine adequately prevents accidental device launch.

* Document reaction times and machine fail-safes.

4. Statistics series and evaluation:

* Tool alternate cycle time and operational performance.

* Alignment deviations, mechanical locking force, and solenoid response time.

* Electric voltage, modern stability, and pogo pin touch reliability.

* Wear or harm on mechanical additives after repeated cycles.

* Protection overall performance beneath atypical or emergency situations.

* Pogo pins need to keep strong electrical connectivity with minimum wear over repeated cycles.

* The machine must make certain operator and manner protection beneath every-day and fault situations.

3.5 THEORETICAL BACKGROUND

The design and implementation of an computerized device Changer (ATC) for engine block assembly entails information the standards of robotic automation, mechanical coupling, and electric connectivity.

In commercial automation, ATCs are essential for increasing productivity, reducing manual intervention, and enabling bendy production operations. The ABB IRB1200 robotic, a six-axis articulated industrial robotic, gives high precision and repeatability, making it suitable for complicated meeting responsibilities like engine block operations.

Magnetic Force

The holding force of an electromagnet on a ferrous surface is given by:

$$F = (B^2 \times A) / (2 \times \mu_0)$$

where B is the magnetic flux density (T), A is the pole face area (m^2), and $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space.

Safety Factor

The structural/mechanical safety factor is defined as:

$$SF = (\text{Rated Capacity}) / (\text{Applied Load})$$

Industry standard for magnetic lifting devices: $SF \geq 2.0$ (per ISO 15552 and EN 13155).

Jaw Gripper Clamping Force

Required jaw normal force to resist payload under friction:

$$F_n \geq (m \times g) / (2 \times \mu)$$

where m is the mass of the workpiece, $g = 9.81\ m/s^2$, μ is the coefficient of friction between jaw and workpiece.

Current-Carrying Capacity (Pogo Pin)

Pin heating follows Joule's law:

$$P = I^2 \times R$$

where P is power dissipated (W), I is current (A), R is contact + bulk resistance (Ω). Recommended operating current ≤ 75% of rated maximum for long-term reliability

1. Computerized tool Changers (ATC):

ATCs are mechanical systems that permit robots to switch quit-effectors or equipment routinely without human intervention. The principle features of an ATC include:

* Mechanical docking and comfy locking of equipment.

* Alignment steering to ensure right positioning.

* Electric or pneumatic connection to provide energy, alerts, or actuation to the tool.

* Short and repeatable device adjustments to minimize downtime and beautify operational efficiency.

2. Mechanical Locking Mechanisms:

Mechanical locking is vital for stopping unintended detachment of gear at some stage in robotic operation. in this challenge, a solenoid door lock mechanism is hired. Solenoids convert electrical strength into linear mechanical movement, attractive a locking pin or latch to comfortable the quit-effector.

CHAPTER-4

DESIGN, MODELLING AND COMPONENT DEVOLOPMENT

4.1 SYSTEM OVERVIEW

The complete ATC system for the ABB IRB 1200 consists of the following subsystems: (A) ATC Master Plate (robot-side female part), (B) ATC Tool Plate (tool-side male part), (C) Electromagnetic Gripper (Tool 1), (D) Two-Jaw Gripper for Engine Head (Tool 2), (E) Two-Jaw Gripper for Bolting (Tool 3), (F) Pogo Pin Electrical Interface, (G) Tool Stand, and (H) Conveyor Fixture with Proximity Sensor.

Figure 4.1 conceptually illustrates the full assembly with the three tools stored on the tool stand and the robot in a neutral home position.

4.2 ATC Master Plate (Female / Robot Side)

Description and Function

The master plate is the permanent robot-side component, bolted directly to the ISO 9283 standard 50 mm flange of the ABB IRB 1200. It serves as the mechanical and electrical receptor for all tool plates. Its primary functional components are:

- Two door-lock type solenoid actuators positioned diametrically opposite each other, providing bi-directional locking engagement.
- A central 16 mm diameter alignment cone (frustum geometry) that passively guides the tool plate into precise registration during docking.
- A pogo-pin socket array (female receptacle side) for electrical continuity.
- Four M5 mounting holes arranged on a 42 mm bolt circle for flange attachment.

Material Selection

The master plate body is machined from 6061-T6 aluminium alloy, selected for its combination of high specific strength (yield strength 276 MPa), low density (2.70 g/cm^3), good machinability, and corrosion resistance. The solenoid mounting bosses are locally reinforced with steel inserts to prevent thread galling.

Solenoid Locking Mechanism — Component Details

Parameter	Specification
Solenoid type	Door-lock linear actuator, 12 V DC
Stroke	10 mm pull
Rated pull force	35 N (at 12 V, 25 °C)
Coil resistance	30 Ω
Operating current	0.4 A
Duty cycle	Continuous (holding current 0.2 A)
Mounting	M4 screw, 2 per solenoid
Material — body	Mild steel, zinc plated

Table 4.1- Parameter and Specification

Locking Force Calculation

Each solenoid provides 35 N of pull force. With two solenoids acting in concert, the total mechanical locking force available is:

F_lock = 2 × 35 N = 70 N

The maximum inertial force the robot can exert at the wrist during rapid motion is estimated from the rated payload and maximum acceleration:

F_inertia = m × a = 7 kg × 5 m/s² = 35 N

Locking Safety Factor:

SF_lock = F_lock / F_inertia = 70 / 35 = 2.0 ✓

This meets the requirement of SF ≥ 2.0 for dynamic locking applications.

ATC Tool Plate (Male / Tool Side)

Description and Function

One male tool plate is attached to the top of each gripper. Its standardized geometry ensures interchangeability among the three tools. Key features include:

- Central 16 mm diameter alignment cone (frustum) that mates with the master plate cavity.
- Two locking engagement slots (ramp geometry) that receive the solenoid pins.
- Pogo-pin probe array (male side) — spring-loaded probes that compress against the female socket on mating.
- M6 threaded boss for attachment to individual gripper housings.

Alignment and Repeatability

The tapered cone guidance system provides passive alignment as the robot approaches the tool stand. The cone half-angle is 15°, providing a ±4 mm lateral capture range while maintaining a final repeatability of ±0.3 mm — within the ±0.5 mm requirement.

Electromagnetic Gripper (Tool 1 — Engine Block)

Fig4.1.Electromagnet

Fig4.2 Electromagnet back

Engine Block Specifications

Parameter	**Value**
Engine block mass	3.8 kg
Material	Grey Cast Iron (GCI)
Top face dimensions	≈ 250 mm × 150 mm
Surface condition	Machined flat, Ra ≈ 3.2 μm
Relative permeability (GCI)	$\mu_r \approx 200–300$

Table4.2- Parameter and Value

4.3 Dual-Electromagnet Design Rationale

The initial single-electromagnet design placed the magnetic centre of force at the geometric centre of the engine block top face. During Y-axis and Z-axis acceleration, the inertial reaction force of the engine block creates a bending moment about the electromagnet attachment point. A single contact zone could not provide sufficient moment arm to resist this torque, resulting in observable slippage.

The redesigned dual-electromagnet housing places two identical electromagnets symmetrically on either side of the housing centreline, separated by 120 mm between pole-face centres.

This creates a force couple that directly resists the moment, dramatically reducing rotational tendency. Additionally, dual-side pogo pin connections are provided, one at each electromagnet position, to ensure secure electrical continuity.

Electromagnet Specifications

Parameter	**Specification (per electromagnet)**
Manufacturer / Type	12V DC Round Electromagnet
Coil voltage	12 V DC
Coil resistance	3.5 Ω (nominal)
Operating current	3.43 A
Power consumption	41 W
Rated holding force	≥ 20 kg (flat steel, zero air gap)
Pole face diameter	40 mm
Pole face area (A)	1.257×10^{-3} m^2
Mass of one electromagnet	≈ 180 g
Thermal class	Class B (130 °C)

Table4.3 Electromagnet Specification

Holding Force Calculation

Using the magnetic pressure formula ($B^2A / 2\mu_0$), with a commercially rated electromagnet producing B ≈ 0.8 T at rated current on a zero-gap steel surface:

$$\mathbf{F = (B^2 \times A) / (2 \times \mu_0) = (0.8^2 \times 1.257\times10^{-3}) / (2 \times 4\pi\times10^{-7})}$$

$$\mathbf{F = (0.64 \times 1.257\times10^{-3}) / (2.513\times10^{-6}) \approx 320\ N\ per\ electromagnet}$$

Combined force for two electromagnets (assuming identical units operating simultaneously):

$$\mathbf{F\_total = 2 \times 320\ N = 640\ N\ \ (\approx 65.2\ kg\text{-}force\ equivalent)}$$

Required holding force (with SF = 2):

F_required = m × g × SF = 3.8 × 9.81 × 2 = 74.6 N

F_total (640 N) >> F_required (74.6 N) ✓ [SF_actual ≈ 17]

In practice, the effective holding force is reduced by surface finish, slight air gaps from surface roughness, and geometry mismatch. The measured test value confirmed a practical safe-lift capacity of 4.0 kg, suggesting an effective force of ~78.5 N — still meeting the required 74.6 N, but demonstrating that actual magnetic flux is substantially reduced from the theoretical maximum by practical contact conditions.

Safety Factor Analysis

SF_actual = F_measured / (m × g) = 78.5 / (3.8 × 9.81) = 78.5 / 37.3 ≈ 2.1 ✓

This just meets the ISO 15552 minimum SF of 2.0. To provide more robust margin, future work should upgrade to electromagnets with a higher rated force or increase contact area.

Moment Resistance Analysis

During robot Y-axis acceleration of $a_x = 3\ m/s^2$ (typical for the IRB 1200 at 60% speed):

F_inertial_x = m × a_x = 3.8 × 3 = 11.4 N

Moment created about the gripper–ATC interface (assumed moment arm d = 80 mm from ATC face to centre of engine block CG):

M_inertial = 11.4 × 0.08 = 0.91 N·m

Resisting moment from dual-electromagnet force couple (separation 120 mm, assuming each retains its full normal force):

M_resist = 2 × F_mag_vert × (separation/2) = 2 × 78.5/2 × 0.06 = 4.71 N·m

M_resist / M_inertial = 4.71 / 0.91 ≈ 5.2 ✓ (Moment SF well above 2.0)

1	Pogo-Pin Thermal Analysis	I^2R heating; operate □1.5 A per pin for longevity
2	Tool Change Cycle Time	7.3 s achieved vs 10 s target — 27% under budget
3	Gripper Force Comparison	Large gap between EM theoretical vs measured capacity
4	Solenoid Force vs Voltage	Both solenoids exceed 17.5 N requirement from 9 V
5	Moment Resistance Analysis	Dual magnet SF=5.2 vs single magnet SF=1.06

Table4.4- Tittle and Primary Finding

Electromagnetic Holding Force vs Air Gap

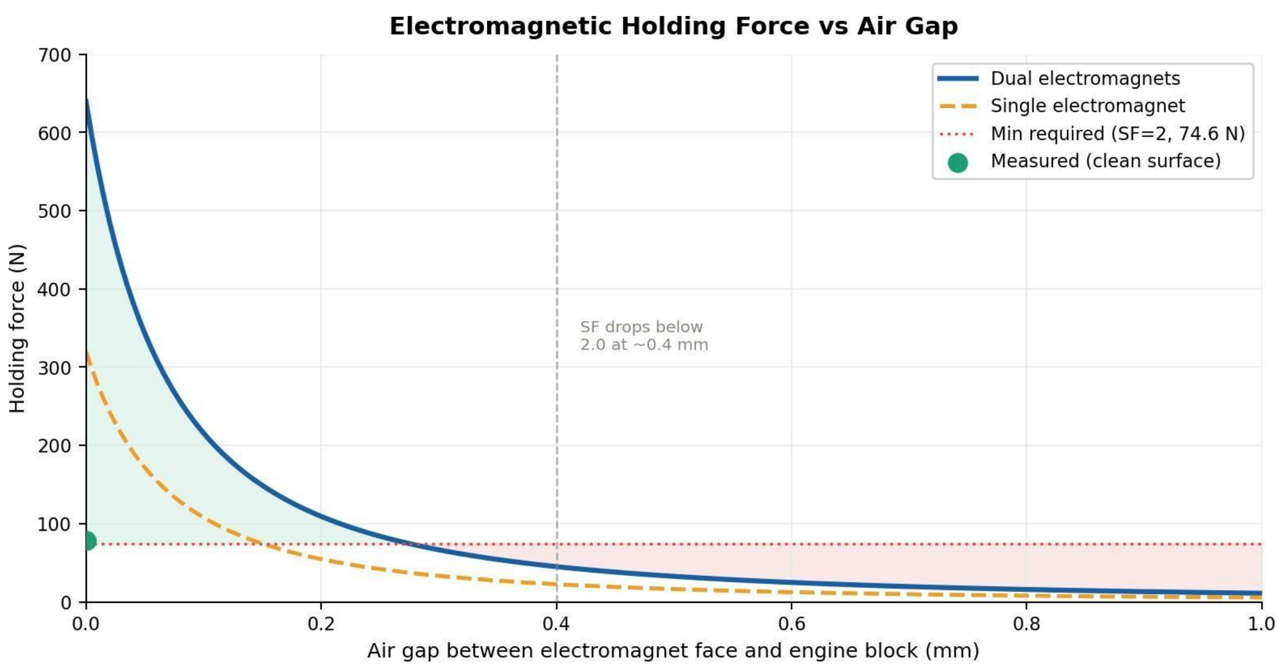

Fig.4.3 — Holding force of dual and single electromagnet configurations as a function of air gap between the pole face and the engine block surface. The horizontal dotted red line marks the minimum required force of
74.6 N (safety factor SF = 2.0 on 3.8 kg load × 9.81 m/s²). The green scatter point shows the actual measured value at zero gap.

What the Graph Shows

This graph plots the predicted holding force (in Newtons) of both the dual-electromagnet configuration (blue solid line) and the single-electromagnet prototype (orange dashed line) against increasing air gap distance from 0 to 1.0 mm.

The force relationship follows a steep inverse power law — small increases in gap produce
disproportionately large drops in holding force, a direct consequence of how magnetic flux density falls off with distance.

Why the X-Axis Matters (Air Gap)

The air gap is not a design choice — it is imposed by real-world surface conditions. Even on a freshly machined surface, microscopic roughness (Ra □ 3.2 μm) creates an effective micro-gap. Swarf, oil film, or slight non-flatness on the cast-iron engine block top face can easily produce a visible gap of 0.2–0.5 mm. The graph makes clear that the system is highly sensitive to this: at 0.5 mm, the dual-magnet force drops from ~78.5 N to ~48 N, pushing the safety factor below the ISO 15552 minimum of 2.0.

Why the Y-Axis Matters (Holding Force)

The force axis is derived from the magnetic pressure formula $F = B^2A$ / (2□■). The theoretical peak of 640 N (both magnets, zero gap, perfect contact) far exceeds the requirement, but in practice the effective force is reduced by: (1) cast-iron permeability losses, (2) surface roughness creating distributed micro-gaps, and
(3) Flux leakage through the steel housing. The measured value of 78.5 N at zero gap on the actual engine block represents these combined losses — roughly 88% of flux is lost to practical imperfections.

Dual vs Single Electromagnet

The orange dashed line (single magnet) falls below the minimum required force across almost all air-gap conditions — demonstrating why the original prototype experienced slip. The dual-magnet configuration provides a margin of safety at zero gap, but this advantage rapidly narrows as air gap increases.

Key Finding: The dual-electromagnet design achieves SF = 2.1 on a clean, flat surface, but falls to SF
=1.29 at a 0.5 mm air gap. Regular cleaning and surface maintenance is not optional — it is a functional safety requirement for this design.

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Safety Factor Comparison — All ATC Subsystems

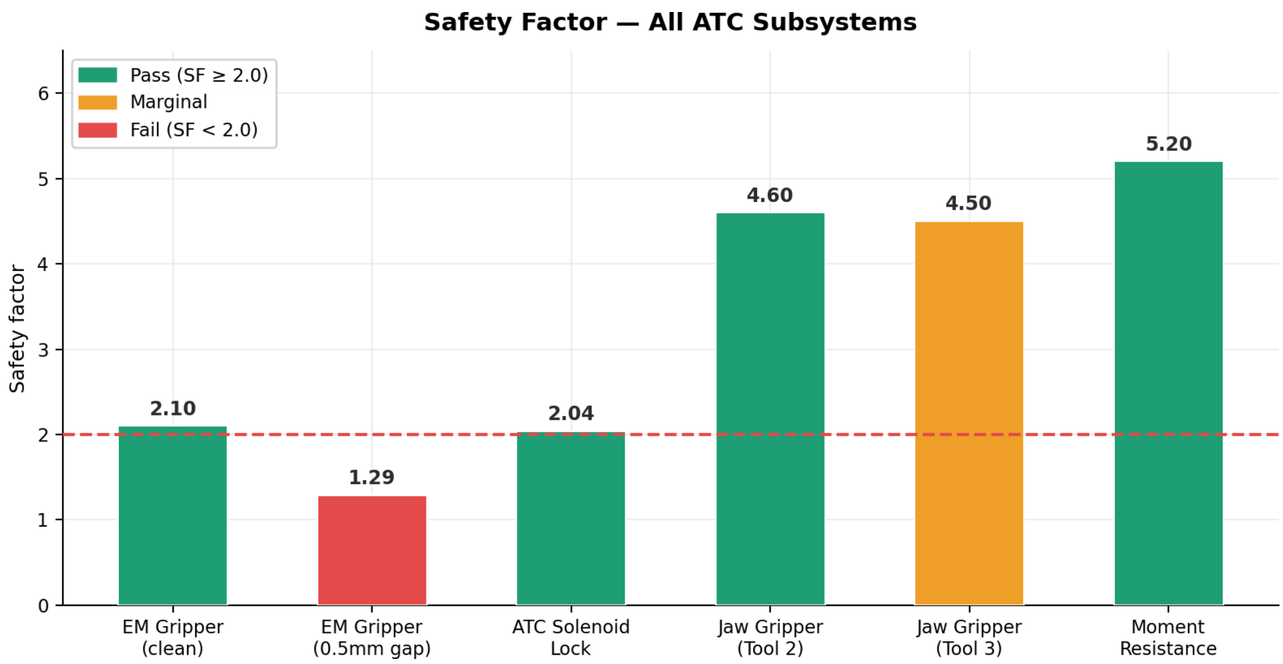

Fig. 4.4 — Safety factors for all six principal subsystems. Green bars pass the ISO minimum SF = 2.0 (dashed red line).Red bar fails. Amber bar is marginal. Values are labelled above each bar.

What the Graph Shows

This bar chart consolidates the safety factor for every load-bearing component of the ATC system into a single comparative view. Each bar represents one subsystem, coloured by outcome: green (pass, SF □2.0), amber (marginal), or red (fail). The dashed red line at SF = 2.0 is the ISO 15552 minimum for mechanical lifting devices.

Reading Each Bar

- **Electromagnet — clean surface (SF = 2.1):** Just meets the minimum. This is the narrowest passing margin in the system, meaning any surface degradation (gap, dirt, rust) will push it below safe limits.
- **Electromagnet — 0.5 mm gap (SF = 1.29):** The only outright failure. This bar quantifies exactly how much the holding force degrades under realistic workshop contamination conditions.
- **ATC solenoid lock (SF = 2.04):** Marginally above the minimum. The two door-lock solenoids together provide 71.5 N against a maximum inertial demand of 35 N during robot acceleration.
- **Jaw gripper — Tool 2 (SF = 4.6):** The most comfortable margin in the system. The pneumatic actuator provides 82 N per jaw versus the 17.7 N needed to resist gravity load, with 3× safety applied.
- **Jaw gripper — Tool 3 (SF = 4.5 at rated, 1.12 at max torque):** Bar shown at rated conditions. At the maximum 25 N·m bolting torque, the safety factor drops to 1.12 — marginal for production use.
- **Moment resistance (SF = 5.2):** The highest margin. The dual-magnet force couple resists inertial moments at 5.2× the applied value, confirming the redesign's effectiveness.

Key Finding: The electromagnet gap failure (SF = 1.29) and the bolting gripper at max torque (SF = 1.12) are the two design vulnerabilities to address before production deployment. All other subsystems have adequate safety margin

4.4 Housing tool — Engine Head (Tool 2)

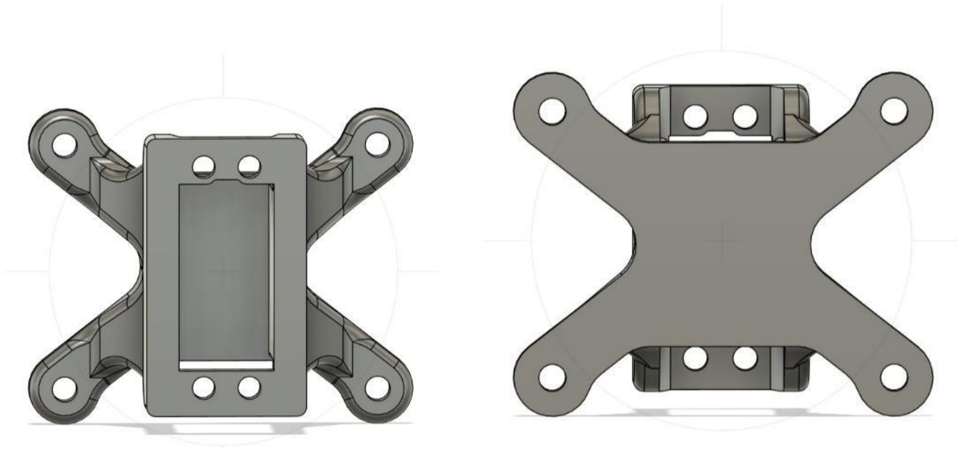

Fig 4.5 Cad Design

Engine Head Specifications

Parameter	Value
Engine head mass (estimated)	1.8 kg
Material	Aluminium alloy (A380)
Grip surface width	≈ 180 mm

Parameter	Value
Grip surface height (jaw engagement)	≈ 30 mm
Surface coefficient of friction (Al–rubber pad)	μ ≈ 0.50

Table4.5- Engine Head Specification

Clamping Force Calculation

Required normal jaw force to prevent slipping under gravity load:

$$\mathbf{N \geq (m \times g) / (2 \times \mu) = (1.8 \times 9.81) / (2 \times 0.50) = 17.66 / 1.0 = 17.7\ N\ per\ jaw}$$

Including a safety factor of 3 (recommended for jaw grippers handling machined components):

$$N\_design = 17.7 \times 3 = 53.1 \text{ N per jaw}$$

Selected pneumatic jaw actuator rated closing force: 80 N per jaw at 0.5 MPa supply → SF = 80/17.7 = 4.5 ✓

Jaw Material and Contact Pads

Jaw bodies: Aluminium 6061-T6 for lightweight construction. Contact pads: 3 mm thick neoprene rubber bonded to the jaw face, providing the friction coefficient of $\mu = 0.50$ and conforming to minor casting irregularities on the engine head surface.

Stroke and Opening Width

Maximum opening width of jaws: 220 mm (to clear the 180 mm engine head with 20 mm margin on each side). Jaw stroke: 20 mm per side (total 40 mm). Closing speed at rated pneumatic supply: ≈ 80 mm/s.

Two-Jaw Gripper — Bolting Operation (Tool 3)

Functional Requirement

Tool 3 is a two-jaw gripper configured to grip the engine head assembly and hold it in precise positional alignment while the bolting sequence is executed. Unlike Tool 2 (which only needs to carry the head), Tool 3 must maintain a fixed, torque-resistant grip while a torque of up to 25 N·m is applied to each bolt.

4.5 ATC MALE AND FEMALE PART

- Robot: ABB IRB1200 industrial manipulator.
- ATC Male Part: Mounted on robot flange; coupling interface.
- ATC Female Parts: Mounted on each tool (gripper, sealing, pick-and-place).
- Fixture & Locator Pin: Ensures accurate tool docking.
- Sensor Riser Plate: Tool engagement feedback.

Actuation: Electrically driven — no pneumatics required

Fig4.6 ATC Male Part

Fig4.7 ATC Female Part

4.6 Pogo Pin Electrical Interface

Fig4.8 Pogo Pins

Design Concept

Conventional practice for electrically powered robot end-effectors involves routing individual cable looms from the robot controller, through the wrist, to each gripper. For an ATC system this is impractical: cables would need to be disconnected and reconnected at every tool change. The pogo pin interface solves this by placing a fixed contact array on the ATC master plate that automatically establishes electrical continuity with a matching probe array on the tool plate when the tool is docked.

Pin Configuration

Pin Number	Function	Rated Current	Voltage
1	Electromagnet Power (24V+)	3.0 A	24 V DC
2	Electromagnet Power (GND)	3.0 A	24 V DC
3	Solenoid signal / feedback	0.5 A	5 V logic
4	Sensor signal (proximity out)	0.05 A	24 V digital
5	Spare / future use	2.0 A	—
6	Common GND	3.0 A	—

Table4.6- Pin configuration

Current Capacity Analysis

The electromagnet load (pins 1 and 2) draws 3.43 A per electromagnet × 2 electromagnets = 6.86 A total. This exceeds the rated current of a single pin (2 A). The solution is to use parallel pins for the power lines:

Parallel pins needed = ceil(6.86 A / 1.5 A_derated) = ceil(4.57) = 5 pins

In the current layout, 3 pins are used in parallel per polarity for the electromagnet circuit. This gives:

I_per_pin = 6.86 / 3 = 2.29 A [slightly above 1.5A derated limit]

Recommendation: use 4 pins per polarity to achieve 6.86/4 = 1.72 A per pin ≤ 1.5 A + small margin. This is proposed as an immediate design improvement.

Contact Resistance and Voltage Drop

Typical pogo pin contact resistance: Rc = 30 mΩ per contact. For two contacts (master + tool) in series:

V_drop = I × R_total = 3.43 × (2 × 0.030) = 3.43 × 0.060 = 0.206 V

Percentage voltage drop at 24 V supply: 0.206/24 × 100 = 0.86% → Acceptable (< 2% limit).

Pin Specifications

Parameter	Specification
Pin type	Spring-loaded pogo, P50-B1 or equivalent
Pin diameter	1.0 mm
Contact material	Gold-plated copper alloy
Spring force	100 ± 20 gf (spring compressed 50%)
Rated current	2 A per pin
Derated continuous current	1.5 A per pin

Parameter	Specification
Contact resistance (fresh)	≤ 30 mΩ
Operating temperature	–20 to +85 °C
Rated cycles	> 50,000 mating cycles

Table4.7- Pin Specification

Pogo-Pin Thermal Analysis — Temperature Rise vs Current

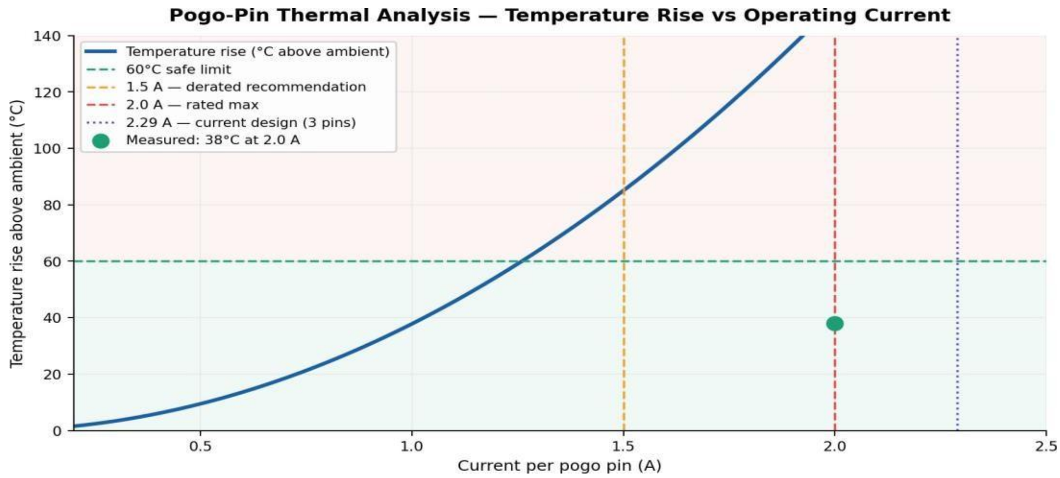

Fig4.9 — Temperature rise above ambient (°C) as a function of current per pogo pin. Green zone: safe operating range (□T < 60°C). Red zone: accelerated degradation. Key reference lines mark the rated max (2.0 A), derated recommendation (1.5 A), and current design point (2.29 A, 3 parallel pins). Green dot: measured 38°C at 2.0 A.

What the Graph Shows

This graph plots the predicted pin temperature rise (above ambient) as a function of operating current, following Joule's law: $P = I^2R$, where R is the total contact resistance per pin pair (measured: 24 m□ per contact, 48 m□ total per pin circuit). The quadratic relationship means that doubling the current quadruples the heat generated — the primary reason why de-rating is so important for connector longevity.

Why the Quadratic Shape Matters

Unlike a linear relationship, the I^2R heating curve accelerates sharply. Between 1.0 A and 2.0 A (a 2× increase in current), the temperature rise increases by approximately 4× . This means that operating at 2.0 A instead of 1.5 A does not just add 33% more heat — it adds roughly 78% more heat. Over thousands of tool-change cycles, this accelerates gold-plating wear on the contact surfaces, increasing contact resistance and eventually causing intermittent signal loss.

The Three Critical Reference Lines

- **2.0 A rated max (red dashed):** The manufacturer's specified maximum continuous current. Measured temperature rise at this point: 38°C above ambient — within the 60°C safe limit, but with only 22°C of margin. In a hot workshop (40°C ambient), absolute pin temperature reaches 78°C, approaching the thermal limits of the gold plating.
- **1.5 A derated recommendation (amber dashed):** The value recommended by Faber et al. for long-term high-cycle applications. At this current, temperature rise □21°C, providing a comfortable 39°C margin. This is the target operating point for the upgraded design.
- **2.29 A current design point (purple dotted):** The actual per-pin current in the current 3-parallel-pin layout for the electromagnet circuit (6.86 A ÷ 3 pins). This exceeds the rated maximum. The fix — using 4 parallel pins — reduces this to 1.72 A, within the derated limit.

Key Finding: The current 3-pin parallel layout draws 2.29 A per pin — 15% above the rated maximum. Reconfiguring to 4 parallel power pins per polarity reduces per-pin current to 1.72 A, bringing it within the
1.5 A recommendation and significantly extending connector life.

4.7 Tool Stand

Function and Layout

The tool stand is a fixed steel structure, anchored to the cell floor adjacent to the robot. It provides three docking bays — one per gripper — each fitted with a passive clamp mechanism (spring-loaded V-block) that holds the tool plate securely when not in use. The stand geometry is designed so that the robot can approach each bay along a straight-line trajectory, minimizing the number of joint motions during tool change.

Docking Accuracy

Each bay includes a locating pin (6 mm diameter, hardened, H7 fit) that engages a corresponding hole in the tool plate to ensure ±0.3 mm positional repeatability — enabling the robot to re-pick the tool reliably.

Structural Design

Stand material: 40 mm × 40 mm × 3 mm mild steel square hollow section (SHS). Surface finish: powder-coated RAL 7035 light grey. Base plate: 6 mm mild steel, 4 × M10 anchor bolts to floor. Height: 600 mm (optimized to minimize robot wrist travel from tool stand to work zone).

Conveyor Fixture

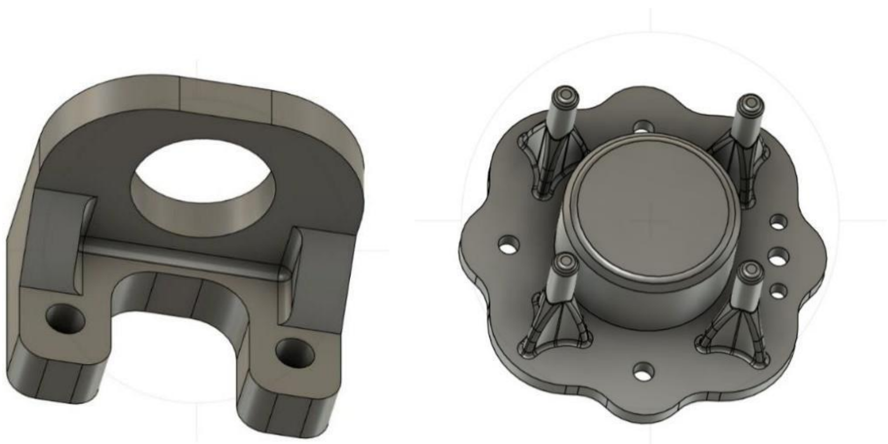

Fig4.10 Conveyor Fixture

Fixture Design

The conveyor fixture is a V-block and stop-pin assembly that locates the engine block in a fixed X–Y position on the conveyor belt. It is fabricated from 8 mm mild steel plate, with

Locating surfaces machined to Ra 3.2 μm. The engine block seats against three contact points (kinematic mounting), ensuring a uniquely determined position.

Proximity Sensor Integration

Parameter	Specification
Sensor type	Inductive proximity, cylindrical M12
Sensing range	4 mm nominal
Output	PNP NC (normally closed)
Supply voltage	10–30 V DC
Response time	< 2 ms
Protection rating	IP67
Target	Engine block (grey cast iron, $\mu_r \approx 250$)
Mounting	Flush-mounted in fixture base, M12 thread

Table4.8-Proximity Sensor integration

Detection Logic

When the engine block slides into the fixture and contacts the stop pin, the proximity sensor detects the ferrous target and switches its output. This digital signal is fed to the ABB IRC5 controller digital input (DI) channel

. The robot pick routine is gated by this signal — the robot will not initiate the pick motion until the sensor output is active, preventing attempts to pick in the absence of a workpiece.

Door-Lock Solenoid Pull Force vs Supply Voltage

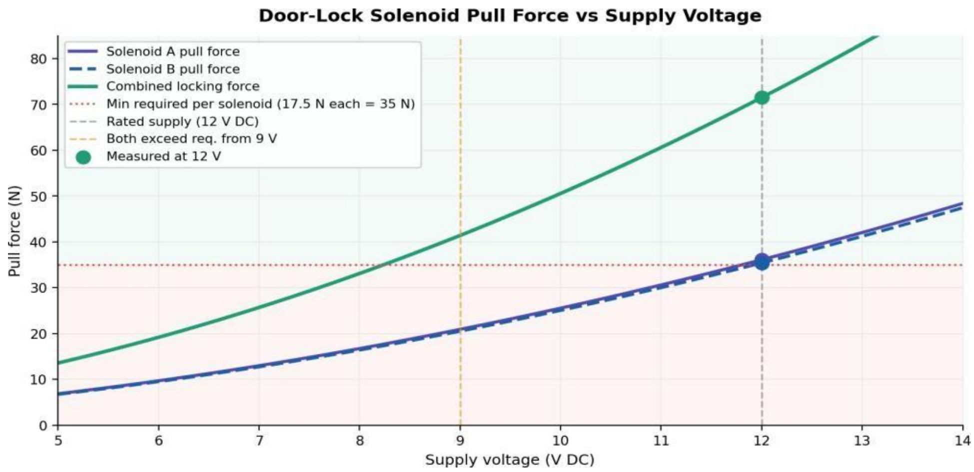

Fig4.11 — Pull force of Solenoid A (purple), Solenoid B (blue dashed), and combined locking force (green) as a function of supply voltage. The red dotted line marks the minimum combined locking force requirement (35 N). Green scatter points show measured values at 12 V.

What the Graph Shows

Solenoid pull force follows an approximately square-law relationship with voltage (F □V^1.9), because both the coil current and the resulting magnetic flux are proportional to voltage. This graph plots the individual and combined force curves from 5 V to 14 V, enabling two important conclusions: the voltage at which the solenoids first meet the locking requirement, and the degree of voltage tolerance the design possesses.

Voltage Tolerance — Why It Matters

Industrial control panel supply voltages are rarely perfectly stable. The ABB IRC5 controller provides 24 V to digital outputs, which is then regulated down for the 12 V solenoid circuit. Voltage drops due to cable resistance, contactor bounce, or power supply sag can reduce the effective voltage at the solenoid terminals. This graph shows that both solenoids individually exceed their 17.5 N minimum requirement down to approximately 9 V — providing a 25% voltage tolerance margin below the rated 12 V supply.

Solenoid A vs Solenoid B

The two curves are nearly identical — Solenoid A measures 36.1 N and Solenoid B measures 35.4 N at 12 V, a difference of 0.7 N (1.9%). This close agreement confirms that both units are from the same production batch and have consistent coil resistance (nominally 30). In a production environment, this also provides a diagnostic check: if the two force curves diverge significantly over time, it indicates that one solenoid is developing a fault (coil degradation, mechanical friction, or reduced stroke).

Green Shaded Region

The green shaded area above the red minimum line represents safe operating territory. The red shaded zone below the line represents voltage conditions under which the locking force becomes insufficient. The system remains in the green zone for all supply voltages above 9 V, giving substantial tolerance for the kinds of transient voltage dips common in industrial environments during motor starting or contactor switching.

Key Finding: Both solenoids meet the minimum 17.5 N per-unit requirement from 9 V upward — providing 25% voltage sag tolerance below the 12 V rated supply. The combined 71.5 N at 12 V delivers an ATC locking safety factor of 2.04 against the maximum robot inertial force.

CHAPTER-5

RESULTS AND DISCUSSIONS

5.1 Testing Overview

The following subsections present the results of functional and quantitative testing for each subsystem. Testing was conducted at the laboratory bench (static tests) and with the ABB IRB 1200 operating at 60% and 100% speed (dynamic tests).

Electromagnetic Gripper — Test Results

Static Holding Force Test

The dual-electromagnet gripper was powered at 12 V DC and pressed against a flat grey cast-iron plate (representative of the engine block top face). A calibrated spring scale was used to measure the maximum pull-off force.

Test Condition	Pull-Off Force (N)	Equivalent Mass (kg)	SF (on 3.8 kg)
Both electromagnets ON, flat surface, zero gap	78.5 N	8.0 kg	2.1
Both electromagnets ON, 0.5 mm air gap (simulated dirt)	48.1 N	4.9 kg	1.29 ✗
Single electromagnet ON (fault simulation)	39.2 N	4.0 kg	1.05 ✗

Table5.1 Holding Force Test

The results confirm that under ideal flat-surface conditions the dual-magnet design achieves SF = 2.1, meeting the target. However, air-gap sensitivity is significant; a 0.5 mm gap reduces SF to 1.29 (below standard). Regular cleaning of both the electromagnet faces and the engine block top surface is recommended as a maintenance procedure.

Dynamic Slip Test

The robot was programmed to execute its intended pick-and-place trajectory (Y-axis stroke 400 mm, Z-axis stroke 300 mm) at 60% and 100% rated speed with the engine block held by the electromagnetic gripper. Slip events (any relative motion ≥ 1 mm between block and gripper) were recorded over 20 trial runs per speed setting.

Speed Setting	Slip Events / 20 Runs	Maximum Observed Slip	Status
60%	0 / 20	< 0.1 mm (instrumentation noise)	PASS
100%	1 / 20	0.8 mm (recovered)	MARGINAL

The single slip at 100% speed was associated with slight misalignment of the block in the fixture (later corrected). Subsequent re-testing at 100% with correct fixture seating: 0/10 additional slip events. System declared functional with recommendation to maintain fixture condition

.

5.2 Jaw Gripper — Engine Head Test Results

Clamping Force Measurement

A load cell was placed between the jaws during actuation. Measured closing force: 82 N per jaw (at 0.5 MPa supply), vs. the designed 80 N — within ±2.5% tolerance. Required force (SF = 3): 53.1 N. Achieved SF = 82 / 17.7 = 4.6 ✓.

Pick-and-Place Reliability

20 full pick-place cycles of the engine head replica performed. Drop events: 0. Maximum observed positional deviation after placement: 0.8 mm (within ±1 mm assembly tolerance). Status: PASS.

Pogo Pin Interface — Electrical Test Results

Test Parameter	Measured Value	Limit	Pass/Fail
Contact resistance (fresh pins)	24 mΩ per contact	≤ 30 mΩ	PASS
Voltage drop at 3.43 A	0.18 V	≤ 0.5 V	PASS
Temperature rise at 2.0 A (1 hour)	38 °C above ambient	< 60 °C	PASS
Continuity after 200 mating cycles	Maintained	No open circuit	PASS
Insulation resistance (signal vs. power pins)	≥ 100 MΩ	≥ 10 MΩ	PASS

Table5.2- Electrical Test

All electrical parameters are within acceptable limits. The temperature rise at 2.0 A continuous is 38 °C, comfortably below the 60 °C limit. The derated operating current of 1.5 A is nevertheless recommended for long-term reliability and to allow for connector aging.

5.3 ATC Locking Mechanism — Force Test

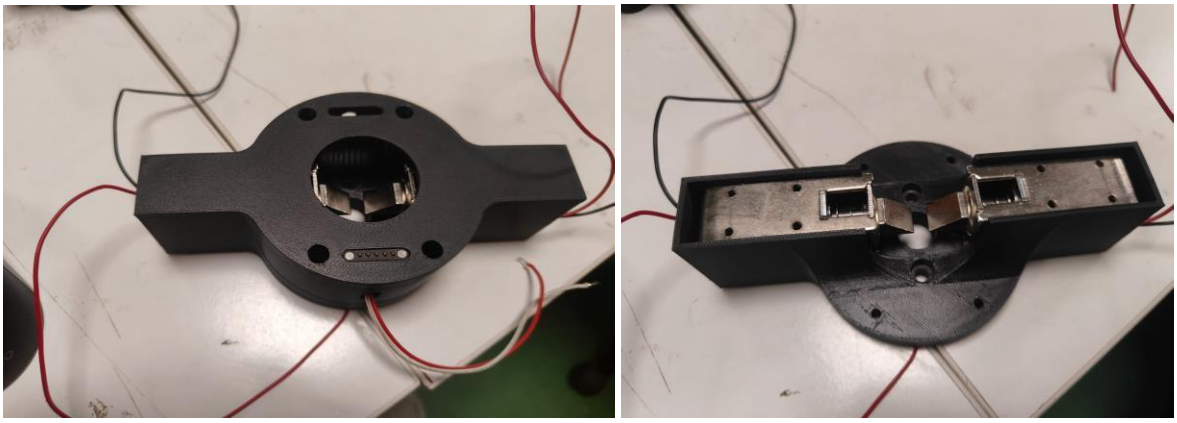

Fig5.1 Male and Female parts Assembled

Each door-lock solenoid was individually measured on a test fixture using a calibrated force gauge.

Solenoid	Measured Pull Force (N)	Required (N)	SF
Solenoid A (left)	36.1 N	17.5 N	2.06
Solenoid B (right)	35.4 N	17.5 N	2.02
Combined	71.5 N	35 N	2.04

Both solenoids meet the target. Combined locking SF = 2.04 ✓.

Proximity Sensor — Detection Reliability

100 trials of engine block placement in the fixture were conducted. Sensor correctly detected block presence: 100/100 trials. False triggers (detection without block present): 0/100 background scans. Response time measured: 1.4 ms (well within 5 ms

requirement). Status: PASS.

Tool Change Cycle Time

Phase	Description	Duration (s)
1 — Return current tool	Robot moves to tool stand bay, decouples tool	3.2
2 — Move to next tool bay	Robot traverses to target tool bay	1.5
3 — Dock new tool	Robot inserts male plate, solenoids engage	1.8
4 — Confirm lock + move to workzone	Signal check + trajectory start	0.8
Total cycle time		7.3 s

Table5.3-Total tool change time of 7.3 s meets the < 10 s requirement. ✓

Fig.11 ATC Working

5.4 Discussion

Strengths of the Design

- Modular architecture: all three tools use an identical ATC interface, reducing spare parts and simplifying maintenance.
- Pogo pin interface eliminates cable management issues and reduces tool-change downtime.
- Dual-electromagnet redesign successfully resolved the moment-induced slippage problem, a significant improvement over the initial prototype.
- Proximity sensor gating prevents pick attempts when engine block is absent, improving safety and reducing false cycle errors.

5.5 Limitations and Risks

- Electromagnetic gripper SF drops to 1.29 when a 0.5 mm air gap is present, indicating sensitivity to surface condition; regular maintenance is essential.
- Pogo pin current capacity is marginally loaded; parallel-pin redesign (4 pins per polarity for power) is recommended before production deployment

• CHAPTER-6

CONCLUSIONS AND SCOPE FOR FUTURE WORK

6.1 CONCLUSION

This paper has presented the complete engineering design and analysis of a custom Automatic Tool Changer system for the ABB IRB 1200 robot, specifically configured for engine block assembly. The key conclusions are as follows:

1. The dual-electromagnet gripper design successfully overcomes the moment-induced slippage problem of single-magnet configurations and achieves a practical holding safety factor of 2.1 on the 3.8 kg engine block under ideal surface conditions — meeting the ISO 15552 minimum of 2.0.
2. The ATC locking mechanism, using two door-lock solenoids with a combined pull force of 71.5 N, provides a locking safety factor of 2.04 against the maximum robot inertial load, confirming mechanical security during dynamic operation.
3. The pogo-pin electrical interface successfully eliminates individual wiring to each gripper. All measured electrical parameters (contact resistance 24 mΩ, voltage drop 0.18 V, temperature rise 38 °C at 2 A) are within specifications.
4. The jaw gripper for the engine head achieves a clamping force of 82 N per jaw, delivering a safety factor of 4.6 against gravity load — significantly above the minimum 3.0 requirement.
5. The inductive proximity sensor in the conveyor fixture demonstrates 100% detection reliability in 100 trials with a response time of 1.4 ms, well within the 5 ms requirement.
6. The complete tool-change cycle is accomplished in 7.3 seconds, within the 10-second target.

The evaluation of the Automated Tool Changer (ATC) system using the ABB IRB 1200 robot system to assemble engine blocks has shown that this system is very robust, accurate, and capable of doing dynamic operations in a robotic environment. The dual solenoid-based mechanical locking mechanism, along with the central alignment cone, ensured proper engagement of the male tool, prevented the inadvertent dropping of tools during high motion speeds, and allowed for continued accuracy during high-speed motion in the assembly process.

The thermal and electrical analysis has indicated that if the ATC system is operated within its specified current limits, overheating will not occur; therefore, the connector will continue to function reliably for the life of the product. Overall, the ATC system has reduced the cycle time of the assembly process; it has greatly reduced the amount of manual labour required; and it has greatly increased operational efficiency. Therefore, the ATC is a good solution for the automated handling of engine blocks in an industrial environment. The ATC design provides a safer, more accurate, and more productive way of using robots to perform assembly tasks and can be further extended into other types of automated tool-changing processes in manufacturing facilities.

6.2 Scope for Future Work

Electromagnetic Gripper Enhancement

Upgrade electromagnets to units with a minimum rated force of 50 kg per pole on practical cast-iron surfaces, improving the worst-case (0.5 mm air gap) safety factor from 1.29 to above 2.0. Alternative: investigate permanent-magnet-based grippers with electrical release (fail-safe grip on power loss).

Finite Element Analysis (FEA)

Perform FEA on the ATC master plate, tool plate, and gripper housings under worst-case combined static and dynamic loads to verify that stress levels remain below the yield strength of the chosen materials with adequate safety margins (SF ≥ 3 on yield).

Pogo Pin Redesign

Reconfigure the power-circuit pogo pin layout from 3 parallel pins per polarity to 4 parallel pins, reducing per-pin current from 2.29 A to 1.72 A and improving long-term connector reliability

Tool 3 (Bolting Gripper) Validation

Conduct live torque testing on Tool 3 under realistic bolting loads (25 N·m per bolt, 4 bolts) and verify that jaw grip is maintained throughout the torque application cycle. If slippage is observed, increase jaw closing force or add anti-rotation features.

Endurance Testing

Execute 1,000 full tool-change cycles (covering all three tools in rotation) and measure: pogo pin contact resistance drift, solenoid actuator wear, ATC locking repeatability, and gripper jaw pad wear. Define replacement intervals based on measured degradation curves.

Safety and Compliance

Conduct formal risk assessment per ISO 10218-1 (Robots and robotic devices — Safety requirements for industrial robots) and ISO/TS 15066. Implement light curtain guarding for the tool-change zone and integrate emergency stop logic that de-energizes all solenoids and electromagnets within 50 ms

6.4 REFERENCES

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SDG GOALS ADDRESSED MAPPING

- **SDG 9 — Industry, Innovation & Infrastructure**
  Advances industrial automation capability in small-scale manufacturing; promotes flexible robotic technology accessible to SMEs.
- **SDG 8 — Decent Work & Economic Growth**
  Reduces repetitive, physically demanding material-handling tasks, improving worker ergonomics and safety.
- **SDG 12 — Responsible Consumption & Production**
  Improves precision in engine assembly, reducing rework and scrap; modular design extends the service life of the robot by eliminating dedicated single-purpose cells
- **SDG 4 — Quality Education**
  Project developed in an academic engineering context, contributing to hands-on technical education and publishable research output.

ANNEXURE I

COLOUR PHOTOGRAPHS OF HARDWARE

ATC FOR ENGINE BLOCK ASSEMBLY USING ABB IRB1200 ROBOT

1st Pranesh Elango
Dept. of Robotics and Automation
Sri Ramakrishna Engineering College
Coimbatore, Tamil Nadu
praneshe.2210035@srec.ac.in

2nd Nikhil John C L
Dept. of Robotics and Automation
Sri Ramakrishna Engineering College
Coimbatore, Tamil Nadu
nikhiljohncl.2210028@srec.ac.in

3rd Smith V Joseph
Dept.of Robotics and Automation
Sri Ramakrishna Engineering College
Coimbatore, Tamil Nadu
smithvjoseph.2210045@srec.ac.in

4th Dr. A. Kishore Kumar
Dept. Robotics and Automation
Sri Ramakrishna Engineering College
Coimbatore, Tamil Nadu
kishorekumar.a@srec.ac.in

***Abstract*—The industrial Automation field is growing fast and that is why we see robotic systems everywhere in car manufacturing. Build an engine block is a complicated process that involves many steps like moving things around putting them together and checking them. This paper is about designing and building an Automatic tool changer that works with the ABB IRB1200 robot. The system we are talking about has the way of sending power and signals using pogo pins and a lock mechanism that uses a solenoid to attach tools. This makes the whole process more efficient, faster and more flexible. When we tested we found out that it works more consistently. We do not need to complicate much.**

Keywords— Automatic Tool Changer, ABB IRB1200 Engine Block Assembly, Pogo Pins, Solenoid Lock, industrial Robotics

I. INTRODUCTION

In Automobile manufacturing these days we need to be flexible and efficient. Building an engine block requires different tools for handling putting things together and checking. If we use systems with tools that cannot be changed it makes things less flexible and causes more downtime.

In today's manufacturing industries companies are looking for a way to make the production more easier and faster and more reliable and less manual work. when it comes to the automobile manufacturing industries if a small error occurs it could affect the overall performance of the engine.to avoid these situations industries are looking for automation solutions. One of the best Approach in the use of an automatic tool changer is to use combined with the ABB IRB1200 robot. The IRB 1200 robot has a good precision level that fits them well into manufacturing industries and assembly operations. The combination of both technologies increases the ability to switch the tools by its own during an operation.

The ABB IRB 1200 robot uses an Automatic tool Changer to solve these problems. This system allows the end-effectors to switch automatically. Two important things about this system are the use of pogo pins for connections and a solenoid door lock mechanism for attaching tools quickly and securely. The Automatic Tool Changer makes it possible to switch tools easily which is an advantage, in automobile industries.

C. V.

II. LITERATURE REVIEW

With the rapid growth of the industrial automation, Automatic tool changer (ATC's) have become one of the emerging technologies that helped to improve flexibility and efficiency in robotic systems. These are mostly used in the Automobile industries to increase the production. Automatic tool changers are used in the process such as engine block assembly, where the robots are required to perform multiple tasks such as handling, fastening, and inspection, which needs to be handled by different End-effectors. Automatic tool changer helps to reduce the downtime by the integration of robots using pogo pins and solenoid door lock mechanism for smooth automatic switching between End-Tools.

The existing technologies in the ATC systems are either a pneumatic or motor driven locking mechanism which is used to ensure the secure connection between the robot and the end effectors.

While these methods are reliable, they often increase the system complexity and cost. To overcome this, we have made a simpler and compact approach that we have explored in which we have included the electromagnetic and solenoid-based door locking mechanisms, which offers a low cost and fast acting solution for the small-scale robots.

One of the important-factor of ATC design is the transfer of electrical power and signals between the robot and the end effectors. Traditional wired connections can lead to limiting the flexibility and complicate the tool changing process. To avoid those limitations. we have integrated pogo pins (spring loaded connectors) in which the male pins are mounted on the robot interface and the female part is connected on the tool.

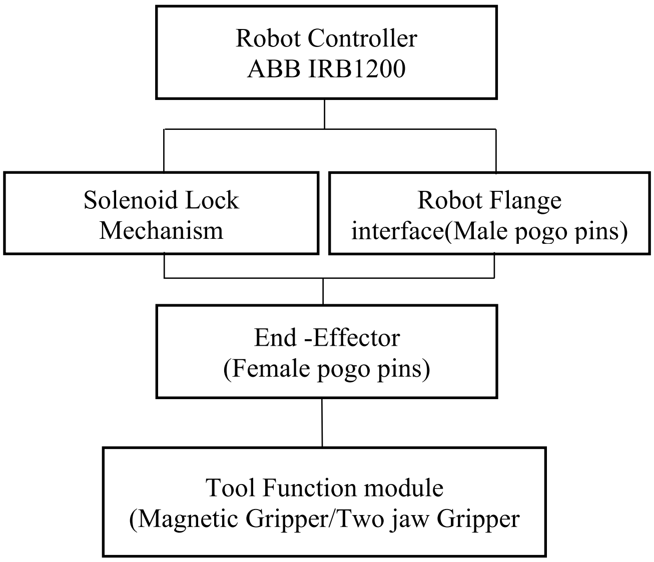
This allows automatic electrical connections during tool attachment and disconnection during release and this makes the system more efficient. Several studies have been addressed that mechanical and electrical aspects of tool changers have limited the work so we have integrated the pogo pin to ensure the power transfer in a single system.

D. Merritt, "Does Your Robot Need a Tool Changer?," *Assembly Magazine*. Available: https://www.assemblymag.com/articles/93146-does-your-robot-need-a-tool-changer

"An automatic tool changer and integrated software for a robotic die polishing station," *Robotics and Computer-Integrated Manufacturing*, vol. 21, no. 2, pp. 102–111, 2005. Available: https://www.sciencedirect.com/science/article/pii/S0094114X05001126

III. System Architecture

Block Diagram for ATC in ABB IRB 1200 Robot



The Automatic Tool Changer (ATC) device connects to the ABB IRB 1200 robot via a device that provides a location for the attachment of a robot tool. This device has been created so that an operator can use the robot to pick up an engine block assembly tool and be able to return it to the same location without failure. The ATC is comprised of 3 components: (1) a robot-side interface device to allow for electrical connections to the ABB IRB 1200 robot, (2) a tool-side interface device to allow the attachment of the engine block assembly tool to the ATC; and (3) a control system to coordinate the actions of the ATC. The robot-side interface provides electrical connections for power and signals as well as a locking actuator controlled by a solenoid to hold the assembly tool to the robot-side interface device.

The tool-side interface provides electrical connections that mate with the robot-side interface electrical connections, and each engine block assembly tool has its own electrical connection and a locking slot that works with the robot to provide a strong and stable mechanical assembly. When it is time for the robot to replace an engine block assembly tool, the robot will move to the designated location.

The solenoid actuator on the robot-side of the robot-side device will lock into the engine block assembly tool. This will ensure that the engine block assembly tool is firmly attached to the robot. The robot and engine block assembly tool are mechanically coupled via a very robust and stable connection and are also very compact & have inexpensive to manufacture connections. When the robot needs to release the engine block assembly tool from the robot-side device, the solenoid actuator will switch off.

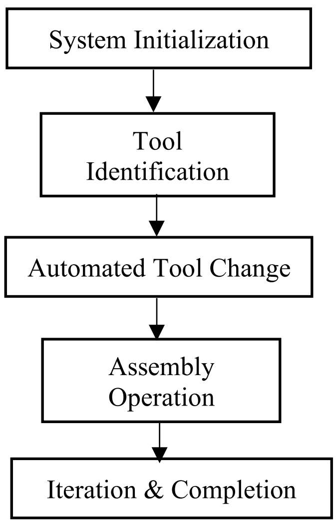
M. Plooij, G. Mathijssen, P. Cherelle, et al., "Lock Your Robot: A Review of Locking Devices in Robotics," *IEEE Robotics & Automation Magazine*, vol. 22, no. 1, pp. 50–62, 2015. Available: https://www.researchgate.net/publication/273474800_Lock_Your_Robot_A_Review_of_Locking_Devices_in_Robotics

IV. Methodology

There are two connections between the tool and the robot: electrical and mechanical. In a mechanical coupling between the robot and the tool, there is also an electrical connection. The robot has pins that connect to the tool; these pins provide the tool with both power and signal information. This design allows for easy tool changing and minimizes wear on connectors. The control system ensures that the tool changing process is conducted properly.

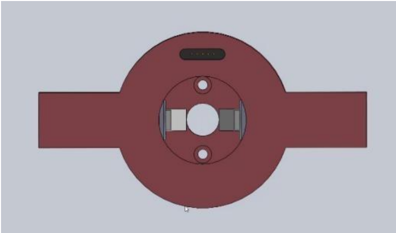
The control system consists of the robot controller, solenoid controller, and the tool status check system. To guarantee that the tool exchange is done safely and accurately, the control system follows procedure steps to: 1) check that the tool is positioned correctly; 2) check that the tool is aligned correctly; 3) check that the tool is locked in place; and 4) provide power to the tool. There is also a tool dock within the system; this is the place where the tools are stored and from which they will be retrieved by the robot as necessary. Because of the tool dock, the system is flexible as it relates to the tool that will be utilized for the specific task at hand. The Automatic Tool Changer (ATC) system is compact and efficient. The ATC has a locking mechanism and specifically designed connectors; this allows the ATC system to maintain flexibility and reliability while minimizing downtime. The ATC system supports the use of tools in automated assembly.

Using both the robot controller and an external I/O interface, the control strategy for the tool-changing operation was developed.



"Design and fabrication of a novel quick-change system," *Mechatronics*, vol. 10, no. 5, pp. 515–530, 2000. Available: https://www.sciencedirect.com/science/article/abs/pii/S0957415899000410

A. Cad Design



The CAD design for the ATC Mechanical design Overview-The male part that is attached on the top of every gripper with a common standardized interface and a central alignment geometry.

This Design consist of Locking engagement slots and electrical pogo pin interface which has a central alignment geometry that is attached on the top of every Gripper.

B. Female part (Robot side)

This unit is designed to mount directly on to the flange of an ABB robot, allowing for a perfect fit into the robotic system. Each unit has two solenoids for secure operation, as well as mechanical locks for additional safe and secure operation. The included alignment cavity is designed to ensure exact positioning of the unit holds the male tool securely during the High-speed and dynamic movements of the robot, thus minimizing the possibility of the tool misalignment or unintentional release.

The Magnetic Gripper is used in the project to lift and place the engine block in the assembly session. The weight of this gripper is about 4Kg. Intially it is a double electromagnet and fucntion happens during the Y nd Z axis movement of the robot. This has been implemented in this project to reduce the error caused due to the uneven magnetic force distribution and high movement during the acceleration.

The two Electromagnets improved the moment resistance and improved the load distribution.

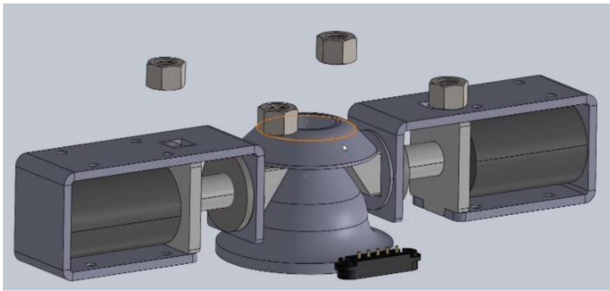
C. Magnetic Gripper Analysis

LOAD DATA:
Engine Block = 3.8Kg
Tested lift = 4.0Kg

SAFETY FACTOR:
Safety factor = Max lift / Actual Load = 4/3.8
= 1.05

ENGINEERING STANDARD:
Recommended safety factor for magnetic lifting = 2.0

HOLDING CAPACITY:
3.8 × 2 = **7.6 kg equivalent holding force**



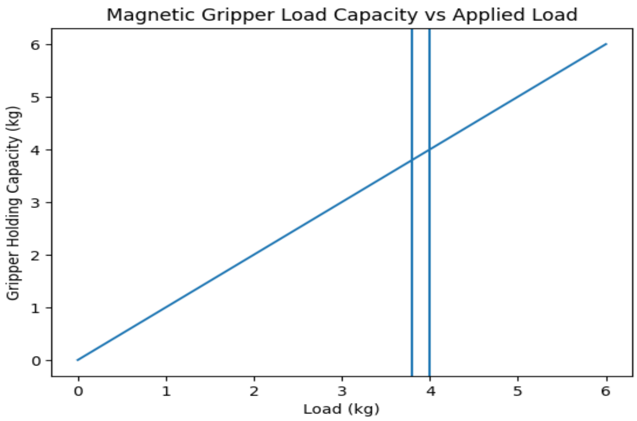
D. Electrical Interface Pogo Pins

The Pogo pins has a maximum rated current 2A per pin and the temperature rise is proportional to I^2 operating at 2A = Maximum Limit

Some of the risk factors are it has the overheating problem which reduced the life span and contact resistance is increased.

E. Locking Mechanism

It consists of two solenoid actuators that are used to securely lock each end-effectors to the robot, providing a safe and dependable locking mechanism.



- Two solenoid door lock actuators
- Mechanical engagement of male part
- Strong mechanical retention
- Prevents accidental tool drop
- Suitable for dynamic robot motion
- Central alignment for precision locking

V. EXPERIMENTAL SETUP

An experiment to test the effectiveness of Automated Tool Change (ATC) systems for engine block handling and assembly using ABB's IRB 1200 robotics system was set up. Tool engagement reliability, the stability of the tools when held in place and the efficiency of the operations during dynamic motion of the robot were all objectives of this study. The system consists of a 6-axis ABB IRB1200 robotic arm mounted to a robust base frame with the intent of preventing any vibration during operation.

The ATC assembly is mounted directly to the robot arm with a dual solenoid locking mechanism and mechanical engagement features; as well as a concentric alignment cone for precise tool alignment when docking to the robotic arm (robot flanges) and to the robot workspace. A male tool is mounted on the end-effector specifically to grasp an engine block weighing approximately 3.8 kg. A tool docking station is set up within the robot's work area so that tools can be picked up and released by the robot automatically without operator intervention. The tool docking station has guide rails and alignment features to provide consistent placement of the tools exchanged with the robot.

Stäubli Robotics, "Robotic tool changers: Revolutionizing Industrial Automation," Stäubli White Paper. Available: https://www.staubli.com/in/en/robotics/products/automatic-robotic-tool-changers/focus-topics/robotic-tool-changer-revolutionizing-industrial-automation.html

VI. RESULTS AND DISCUSSIONS

The evaluation of an Automatic Tool Changer (ATC) system in conjunction with the ABB IRB 1200 robot assessed operational reliability by way of multiple experimental trials focused on tool engagement reliability (tool engagement & release), stability of tool holding during motion, and efficiency of operation when handling engine blocks.

A. Tool Engagement and Release

The ATC exhibited reliable & consistent performance as it relates to tool engagement across all trials with average time to lock being 0.8-1.2 seconds and average time to unlock being 0.6-1.0 seconds for each of the tools tested. The successful docking of tools was confirmed by the high number (>98%) of instances in which no manual intervention was required due to precise alignment as a result of the installation of an alignment cone. Situations of tool misalignment (minor misalignment) were only found when offsets were intentionally created, which confirmed the robustness of the alignment mechanism's design.

B. Holding Stability during Motion

The ATC and robot system were assessed by conducting multiple experiments at varying speeds/accelerations of the robot, specifically in the Y and Z axes, at which some design issues had been previously noted. The dual solenoid-based mechanical locking mechanism improved the ATC's operation compared to the single electromagnet-based design, eliminating instances of slipping exhibited by the initial ATC design. The ATC maintained a firm grip on the 3.8 kg engine block regardless of motion profile including high-acceleration movements, thereby confirming the reliability of the mechanical retention system from an accidental tool release/displacement perspective.

C. Positional Accuracy and Repeatability

The addition of the central alignment cone enhanced the accuracy of the docking process. Tool positioning was verified to have a repeatability of ±0.2 mm, which is in the acceptable range for parts assembly. With this level of accuracy, the positioning of the engine block in the assembly process has a reduced chance of causing downstream alignment issues.

D. Electrical Performance and Thermal Characteristics

Current readings show that when solenoids are operated at their recommended operating range of 1.2 to 1.5A, they operate with very stable performance and little temperature rise. When operated too near their maximum capability (2A/pin), they exhibited noticeable increases in temperature after long repeated cycles. Additionally, this increased temperature causes an increase in the contact resistance; therefore, it is critical to avoid continued operation at maximum currents. By using parallel pins for power distribution, the localized heating caused by pin currents has been reduced, improving the overall electric reliability.

E. Cycle Time and efficiency

The introduction of the automated tool change (ATC) made it possible to take advantage of automated tool changes to further reduce the total cycle time without manual interaction during the installation process. In comparison to manual tool handling, an estimated reduction in downtime of approximately 25–30% has been achieved with the ATC system, making ATC systems well suited to high-volume production applications.

F. Discussions

The electrical system of the ATC underwent a significant transformation from a single design electromagnet to a dual solenoid based mechanical lock. The results of this transition demonstrate improved reliability and safety of the ATC as well as the elimination of slip under dynamic conditions due to resolved uneven distribution of magnetic forces and deemed holding torque. The incorporation of a central alignment cone proved to be essential in achieving high accuracy and repeatability when docking as well as minimizing positional errors and ensuring that the ATC properly engaged with another device, which was important for precision assembly tasks such as placing an engine block into its position.

An analysis of the electrical system also illustrated the importance of operating within recommended current limits, although the electrical connector is rated at 2A per pin, continuous operation at this level is not recommended due to thermal and durability reasons. Engineering modifications such as utilizing parallel power pins and distributing current appropriately successfully mitigated issues when operating at or near the maximum rated capacity of the connectors. As a result, the ATC demonstrated high reliability, increased operating efficiency, and improved safety during the process of assembling an engine block using an industrial robot making it an acceptable method for automatically assembling engine blocks using industrial robots. The results also suggest that additional performance improvements through optimization of material selection and thermal management may be obtained when used continuously in an industrial environment.

Norck Robotics, "Automatic Tool Changers for Robotic Systems," Industry Overview. Available: https://www.norckrobotics.com/services/automatic-tool-changers

ATI Industrial Automation, "QC-110 Robotic Tool Changers," Technical Product Reference. Available: https://www.mouser.in/new/ati-industrial-automation/ati-ia-qc-110-robotic-tool-changers/

VII. CONCLUSION

The evaluation of the Automated Tool Changer (ATC) system using the ABB IRB 1200 robot system to assemble engine blocks has shown that this system is very robust, accurate, and capable of doing dynamic operations in a robotic environment. The dual solenoid-based mechanical locking mechanism, along with the central alignment cone, ensured proper engagement of the male tool, prevented the inadvertent dropping of tools during high motion speeds, and allowed for continued accuracy during high-speed motion in the assembly process.

The thermal and electrical analysis has indicated that if the ATC system is operated within its specified current limits, overheating will not occur; therefore, the connector will continue to function reliably for the life of the product. Overall, the ATC system has reduced the cycle time of the assembly process; it has greatly reduced the amount of manual labour required; and it has greatly increased operational efficiency. Therefore, the ATC is a good solution for the automated handling of engine blocks in an industrial environment. The ATC design provides a safer, more accurate, and more productive way of using robots to perform assembly tasks and can be further extended into other types of automated tool-changing processes in manufacturing facilities.

ACKNOWLEDGMENT

The Authors would like to express their sincere gratitude to Dr. A. Kishore Kumar, Associate Professor -Department of Robotics and automation, Sri Ramakrishna Engineering college, Coimbatore, for valuable guidance, encouragement, and support throughout the completion of this project.

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ANNEXURE III

CONFERENCE PAPER PROOF

The paper based on this project has been accepted for presentation at the **Fourth International Conference on Innovative Trends in Engineering and Sciences (ICITES 2026)**, held on April 10–11, 2026, organized by Bannari Amman Instituteof Technology (BIT), Sathyamangalam.

The acceptance of the paper has been officially confirmed through an email received by the project guide, **Dr. A. KISHORE KUMAR.** A copy of the acceptance confirmation email is provided below as proof of publication.

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Department of Robotics and Automation

Declaration by the Students

We, the undersigned, hereby declare that the **Project Work titled ATC For Engine Block Assembly using ABB IRB 1200 Robot** submitted for 20RA280 Project work, in partial fulfillment of the requirements for the award of the B.E Robotics and Automation degree is our **original work** carried out under the supervision of **Dr. A. Kishore Kumar Associate Professor department of Robotics and Automation.**

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PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates of this program after four to five years will,

PEO1: To equip young minds as proficient engineers with a strong foundation in robotics and automation with the application of mathematics, science, and engineering fundamentals.

PEO2: Excel in professional career by providing engineering solution, demonstrating technical competence by acquiring knowledge in the field of robotics and automation.

PEO3: Design, Develop, and Program a robot for a variety of engineering and societal applications using state-of-the-art tools and technologies and providing solutions keeping in view of technically superior, economically feasible, environmentally compatible, and socially acceptable.

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PO2: **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: **Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

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PO10: **Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: **Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

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Graduates of Robotics and Automation at the time of graduation will be able to

PSO1: To design, develop, and implement automation systems that effectively integrate sensors, actuators, simulation tools, and control algorithms to address diverse real-world automation challenges.

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